Computer Vision Practical Report:

VIDEO(SLOW/FAST)MOTION

1. AIM

To read an input video using OpenCV, change its playback speed to create slow-motion or fast-motion video, and save the processed video to disk.

2. PROCEDURE

- Load the input video using cv2.VideoCapture().
- Read the video properties such as frame rate, width, and height.
- Change the frame rate to control the video speed.
- For **slow motion**, decrease the frame rate or repeat frames.
- For **fast motion**, increase the frame rate or skip frames.
- Create an output video using cv2.VideoWriter().
- Read and process each frame using cap.read().
- Write the processed frames to the output video using out.write().
- Release the video objects using cap.release() and out.release().
- Save the resulting video as **slow_motion.mp4** or **fast_motion.mp4**.

3. PROGRAM CODE

```
import os
import cv2
import matplotlib.pyplot as plt

folder_path = r"C:\Users\adminuser\Desktop\Python CV"

video_files = [f for f in os.listdir(folder_path) if f.lower().endswith(('.mp4', '.avi', '.mkv', '.webm', '.mov'))]

if not video_files:
    print(f"No video files found in {folder_path}!")
```

```python
    print("Files in folder:", os.listdir(folder_path))
    exit()

video_path = os.path.join(folder_path, video_files[0])
print(f"Opening video: {video_path}")

cap = cv2.VideoCapture(video_path)

if not cap.isOpened():
    print("Error: OpenCV still cannot open the video file.")
    exit()

print("1 - Fast Motion")
print("2 - Slow Motion")
choice = input("Enter your choice (1 or 2): ")

step = 10 if choice == "1" else 1
pause_time = 0.001 if choice == "1" else 0.1

plt.ion()
fig, ax = plt.subplots(figsize=(8, 5))

frame_count = 0
ret, frame = cap.read()

if ret:
    frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    im = ax.imshow(frame_rgb)
    ax.axis('off')
    ax.set_title("Fast Motion" if choice == "1" else "Slow Motion")

while ret:
    if frame_count % step == 0:
        frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
        im.set_data(frame_rgb)
```

```
        fig.canvas.draw_idle()
        plt.pause(pause_time)

    frame_count += 1
    ret, frame = cap.read()

cap.release()
plt.ioff()
plt.show(block=True)
```

4. OUTPUT

Slow Motion

Fast Motion

5. RESULT

The Python script executed successfully. The input image 'E6-IMG.png' was loaded.